

Notes: Kontoghiorghes (JFE, 2024)

Do personal taxes affect investment decisions and stock returns?

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1 Big picture

- **Question.** Do personal investment taxes affect stock returns and firms' real decisions?
- **Natural experiment.** In 2013 the UK allowed stocks listed on the Alternative Investment Market (AIM) to be held in tax-exempt Individual Savings Accounts (ISAs) for the first time.
- **Treatment.** AIM stocks moved from being exposed to capital gains and dividend taxes for ISA investors to being exempt. Main-market LSE/FTSE stocks were already ISA-eligible, so they serve as a natural control group.
- **Main stock-return result.** Excess AIM stock returns fell after the reform by about the pre-reform effective tax rate, roughly 0.9 percentage points per month.
- **Tax capitalization interpretation.** Before the reform, AIM investors required compensation for personal tax liabilities. Once AIM stocks became ISA-eligible, this tax premium was capitalized away through higher prices and lower subsequent expected returns.
- **Main firm-decision result.** AIM firms increased dividends, share issuance, debt, capital investment, total pay, and employment after the tax cut.
- **Traditional view versus new view.** The evidence supports the traditional view that personal taxes affect firms' cost of capital and investment decisions, rather than the new view that retained earnings alone determine payout and investment.
- **Heterogeneity.** Non-dividend payers and firms issuing more outside capital respond more strongly in real investment; concentrated-ownership firms increase pay more.
- **Mechanism.** The reform expanded tax-exempt retail demand, lowered AIM firms' equity cost of capital, raised stock prices, and made external financing and real investment more attractive.
- **Economic takeaway.** Personal taxes do not just redistribute investor after-tax income. In this setting, they affect asset prices, expected returns, capital structure, and real corporate investment.

2 Data and measurement

2.1 Institutional setting: AIM, LSE, and ISAs

- AIM is a sub-market of the London Stock Exchange designed for smaller and growing companies.
- The main LSE stocks, represented by FTSE All-Share constituents, had always been eligible for Stocks and Shares ISAs.
- Stocks and Shares ISAs exempt qualifying investments from capital gains tax and dividend tax.
- On July 3, 2013, the UK Treasury announced that AIM stocks would become ISA-eligible from August 5, 2013.
- The stated policy goal was to provide smaller growth firms access to a large pool of "patient capital."

Interpretation. The reform changes the tax treatment of one group of publicly listed firms while leaving otherwise comparable listed firms unaffected. That makes it useful for causal inference.

2.2 Treatment and control groups

- The treated sample is the AIM 100 sample.
- The main control group is a propensity-score-matched FTSE sample.
- Matching covariates include size, age, sector, book-to-market, and market beta.
- The full FTSE All-Share sample is also used in robustness and return-spread tests.

Interpretation. The matched control group is designed to satisfy parallel trends by comparing AIM firms to main-market firms with similar pre-reform observables.

2.3 Stock-return data

- The return analysis uses monthly stock returns from 2004 to 2022.
- Excess returns are analyzed using factor-model regressions and difference-in-differences specifications.
- Factor controls include market, size, value, profitability, investment, momentum, and liquidity factors.
- The paper studies AIM returns alone, AIM minus matched FTSE returns, and AIM minus full FTSE returns.

Interpretation. The return tests ask whether the pre-reform tax premium disappears when AIM stocks become tax-exempt inside ISAs.

2.4 Firm financial decision data

- Firm outcomes include dividends, a dividend dummy, log shares issued, debt relative to pre-reform assets, paid-in equity, cash, property plant and equipment, capital expenditure, total pay, and employment.
- Most firm outcomes are scaled by pre-reform 2012 assets, revenue, or PPE to keep the treatment effect interpretable and reduce endogeneity from contemporaneous denominators.
- The analysis covers 2005-2022 for firm outcomes.

Interpretation. The outcome set separates financial-policy responses from real investment responses. This is important because the theory predicts both: lower equity cost of capital should change financing and investment.

2.5 Effective tax rate

The average effective tax rate for AIM stocks before the reform is estimated as

$$\hat{\tau} = \frac{1}{T} \sum_{t=1}^T \left[\frac{1}{I} \sum_{i=1}^I (\tau_{c,t} R_{i,t} + \tau_{d,t} DY_{i,t}) \right] = 10.7\% \text{ per year}, \quad (1)$$

which is about 0.9% per month.

- $\tau_{c,t}$ is the capital gains tax rate.
- $R_{i,t}$ is annual net stock return.
- $\tau_{d,t}$ is the dividend tax rate.
- $DY_{i,t}$ is the dividend yield.

Interpretation. This effective tax rate is the quantitative benchmark. If tax capitalization holds, the return decline after the reform should be close to this tax premium.

3 Models and empirical specifications

3.1 Tax capitalization equation

The paper starts from a Brennan-style asset-pricing relation:

$$E(R_i - R_f) = \tau_i + \beta'_i X, \quad (2)$$

where τ_i is the tax premium and X contains priced risk factors.

Interpretation. If investors require compensation for personal tax liabilities, pretax expected returns should be higher when taxes are higher. When AIM stocks become ISA-eligible, τ_i should fall toward zero.

3.2 Portfolio factor-model test

The empirical version is

$$\bar{R}_{i,t} - R_{f,t} = \alpha_i + \beta'_{1,i} X_t + \mathbf{1}_{t \geq 7/13} (\gamma_i + \beta'_{2,i} X_t) + \epsilon_{i,t}. \quad (3)$$

- α_i captures the pre-reform tax premium.
- γ_i captures the post-reform change in expected returns.
- Tax capitalization predicts $\alpha_i \approx 0.9\%$ and $\gamma_i \approx -0.9\%$ per month.

Interpretation. The identifying logic is direct: if AIM investors no longer need pretax compensation for personal taxes, the after-reform coefficient should remove the pre-reform tax premium.

3.3 Monthly stock-return difference-in-differences

The main return DID is

$$R_{i,t} - R_{f,t} = \alpha_i + \alpha_{s,t} + \beta \text{Treated}_i \times \text{Post}_t + \gamma' \text{Controls}_{i,t} + \epsilon_{i,t}. \quad (4)$$

- α_i are firm fixed effects.
- $\alpha_{s,t}$ are sector-by-month fixed effects.
- Treated_i indicates AIM stocks.
- Post_t begins in July 2013.
- Controls include risk factors and stock-level proxies for liquidity and information.

Interpretation. This specification uses individual stock observations and absorbs firm and time-varying sector shocks. The coefficient β measures how AIM excess returns changed relative to FTSE controls.

3.4 Dynamic difference-in-differences

The annual dynamic event-study specification is

$$R_{i,t} - R_{f,t} = \alpha_i + \alpha_{s,t} + \sum_{k=-k_0, k \neq 0}^T \beta_k \text{Treated}_i \times \mathbf{1}(k = t) + \epsilon_{i,t}, \quad (5)$$

where 2012 is the omitted baseline year.

Interpretation. Event studies test whether AIM and control outcomes followed parallel pre-trends and whether the post-reform effect is persistent rather than transitory.

3.5 Firm-decision regressions

For financial and real outcomes, the paper estimates

$$Y_{i,t} = \alpha_i + \alpha_{s,t} + \beta \text{Treated}_i \times \text{Post}_t + \gamma' \text{Controls}_{i,t} + \epsilon_{i,t}. \quad (6)$$

Outcomes $Y_{i,t}$ include dividends, shares issued, debt, PPE, capex, pay, employment, and cash.

Interpretation. If the personal tax cut lowers AIM firms' equity cost of capital, then firms should find it cheaper to finance investment and should increase real spending, not only distribute cash to investors.

4 Identification and empirical design

4.1 Source of variation

- The 2013 reform changes the tax treatment of AIM stocks, but not main-market LSE stocks.
- The reform affects the tax treatment of investors, not firms' product markets or technologies directly.
- The policy was designed around exchange listing eligibility, creating a clean treated-control contrast.

Interpretation. The experiment isolates personal investment taxes from many macroeconomic forces that normally move taxes and asset prices at the same time.

4.2 Parallel trends and matching

- Propensity-score matching makes AIM and matched FTSE firms similar on pre-reform observables.
- Firm fixed effects absorb time-invariant firm differences.
- Sector-time fixed effects absorb sector-level shocks.
- Dynamic DID plots check for pre-trends.

Interpretation. The paper does not need AIM and FTSE firms to be identical. It needs them to have followed similar outcome trends absent the reform.

4.3 Why this setting is informative

- The tax change is large: AIM stocks move to zero capital gains and dividend taxation inside ISAs.
- The change is likely perceived as permanent because it equalizes AIM and main-market treatment.
- ISAs are important in the UK retail-investor market and contained over £200 billion in investments in 2013.
- The reform creates cross-sectional and time-series variation in tax treatment.

Interpretation. The effect should show up in prices if retail investors are marginal or if their demand is large enough to affect equilibrium prices.

4.4 Threats and robustness

- **Anticipation:** there was discussion of AIM ISA eligibility before the announcement, but the exact timing was not known and the implementation window was short.
- **Other AIM-specific shocks:** the paper argues no other 2013 event affected only AIM stocks or ISAs.
- **Stamp duty:** AIM stamp duty was removed in 2014, but the tax was small relative to monthly returns.
- **Control group mismatch:** the paper uses matched FTSE controls, full FTSE controls, factor controls, and robustness tables.
- **Exchange switching:** appendix evidence shows switching between AIM and main-market LSE does not drive the results.

Interpretation. The identification is strongest for returns because the reform directly changes the tax wedge. Firm decision results require the additional interpretation that lower cost of capital changes corporate behavior.

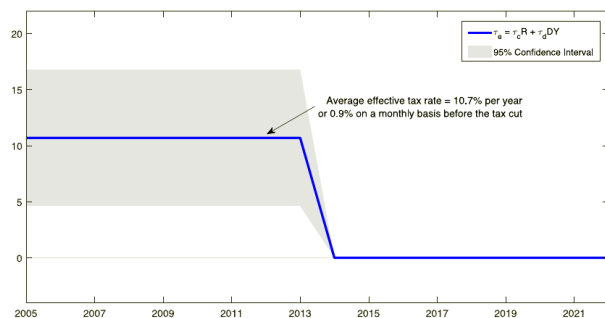
5 Tables and figures

5.1 Figure 1 and Table 1: the tax treatment and matched sample

Read: Figure 1: AIM stocks faced an effective personal tax rate of about 10.7% per year, or 0.9% per month, before 2013. After ISA eligibility, the effective tax rate for ISA investors falls to zero.

Read: Table 1: The AIM 100 is very different from the full FTSE sample, especially in size, age, beta, ROE, and volume. The propensity-score matched FTSE sample is much closer: differences between AIM and matched FTSE firms are statistically insignificant across the displayed characteristics.

Economic message. The reform creates a large tax shock, and the matched FTSE group provides a credible comparison group for treated AIM firms.



(a) Figure 1: AIM effective personal tax rate

Standard errors are in parentheses below sample averages. Stars denote whether sample averages are statistically different from zero, *p < 0.1.

Characteristic	Sample Averages			Sample Differences	
	AIM	Matched FTSE	Full FTSE	AIM/Matched	AIM/Full
Log Size	18.86 (0.08)	19.02 (0.08)	20.22 (0.06)	0.17	1.36***
Age	11.90 (1.19)	14.29 (1.21)	27.10 (1.20)	2.38	15.20***
Value (P/B)	1.37 (0.17)	1.65 (0.23)	1.07 (0.07)	0.28	0.30
Beta	0.38 (0.04)	0.42 (0.04)	0.68 (0.02)	0.04	0.30***
Dimson Beta	0.72 (0.05)	0.77 (0.04)	0.87 (0.02)	0.05	0.15***
Price/Earnings	32.02 (5.02)	25.55 (4.10)	25.19 (1.56)	6.47	6.84
ROE	6.76 (2.44)	7.60 (2.10)	14.32 (1.04)	0.85	7.57**
Log Volume	12.20 (0.19)	12.44 (0.13)	12.81 (0.07)	0.23	0.61***

(b) Table 1: sample averages

5.2 Tables 2-3 and Figure 2: stock-return capitalization

Read: Table 2:

- Before the reform, the AIM portfolio has a positive alpha around 1.1%-1.4% per month depending on factor controls.
- After the reform, the post coefficient is around -0.8% to -1.4% per month.
- The data do not reject the tax capitalization null values of $\alpha = 0.9\%$ and $\gamma = -0.9\%$.

Read: Table 3:

- In the individual-stock DID, AIM monthly returns fall by 1.1 percentage points relative to matched FTSE controls.
- The estimate is very close to the 0.9 percentage point effective tax rate.
- High pre-reform-tax AIM stocks fall more than low-tax AIM stocks, reinforcing the tax-premium interpretation.

Read: Figure 2: The dynamic return estimates show no strong pre-trend and a persistent post-2013 decline in AIM excess returns relative to both matched and unmatched FTSE controls.

Economic message. Returns behave as if the personal tax premium was capitalized into prices: after the tax cut, prices rise and future expected returns fall by about the size of the eliminated tax burden.

$$R_{i,t} - R_{f,t} = \alpha_i + \beta'_{1,t} X_t + \mathbb{I}_{(t \geq 7.13)} (\gamma_i + \beta'_{2,t} X_t) + \varepsilon_{i,t}$$

Panel A: X = Market Factor		$H_0: \alpha_i = 0.9\%, \gamma_i = -0.9\%$	
i	AIM	AIM - Matched FTSE	AIM - Full FTSE
α_i	0.013 (0.005)	0.008 (0.003)	0.005 (0.003)
γ_i	-0.014 (0.005)	-0.012 (0.003)	-0.009 (0.004)
Adj R-squared	0.49	0.07	0.04
Panel B: X = Fama & French 3 Factors		$H_0: \alpha_i = 0.9\%, \gamma_i = -0.9\%$	
α_i	0.011 (0.004)	0.007 (0.003)	0.004* (0.003)
γ_i	-0.013 (0.004)	-0.011 (0.003)	-0.008 (0.004)
Adj R-squared	0.66	0.11	0.19
Panel C: X = All Factors		$H_0: \alpha_i = 0.9\%, \gamma_i = -0.9\%$	
α_i	0.014* (0.003)	0.007 (0.003)	0.005 (0.003)
γ_i	-0.014 (0.004)	-0.010 (0.004)	-0.008 (0.004)
Adj R-squared	0.69	0.13	0.22

(a) Table 2: factor-model return tests

	Dependent Variable: Monthly Return			
	$H_0: \beta = -0.9\%$	$H_0: \beta = -1.9\%$	$H_0: \beta = -0.03\%$	
	(1)	(2)	(3)	(4)
Treated $\times$ Post	-0.011 (0.004)	-0.009 (0.003)	-0.021 (0.007)	-0.002 (0.006)
Fixed Effects	Yes	Yes	Yes	Yes
Clustered SE's	Yes	Yes	Yes	Yes
All Controls	Yes	Yes	Yes	Yes
Months	210	210	210	210
Observations	23,364	85,975	77,872	77,120
Adj R-squared	0.19	0.28	0.30	0.30

(b) Table 3: monthly return DID

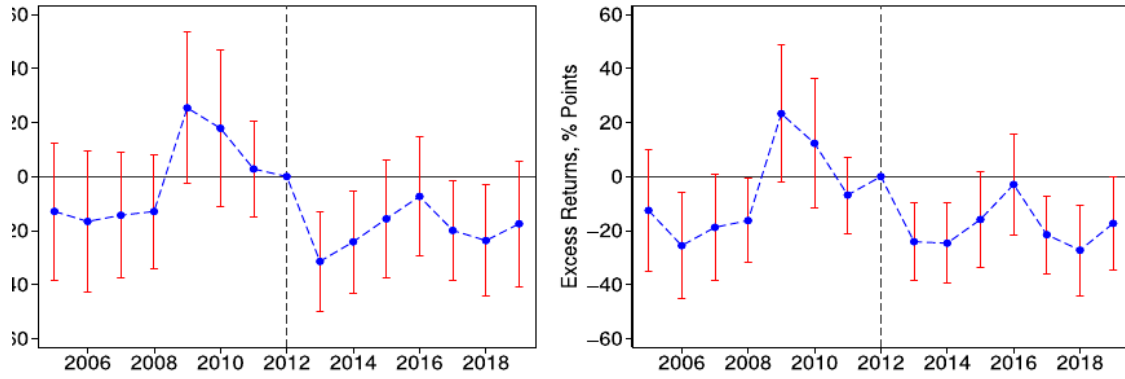


Figure 2: dynamic DID for excess AIM returns

5.3 Table 4 and Figure 3: payout and capital structure

Read: Table 4:

- Log dividends rise by about 40% with controls.
- The dividend dummy rises, but the extensive-margin effect is weaker.

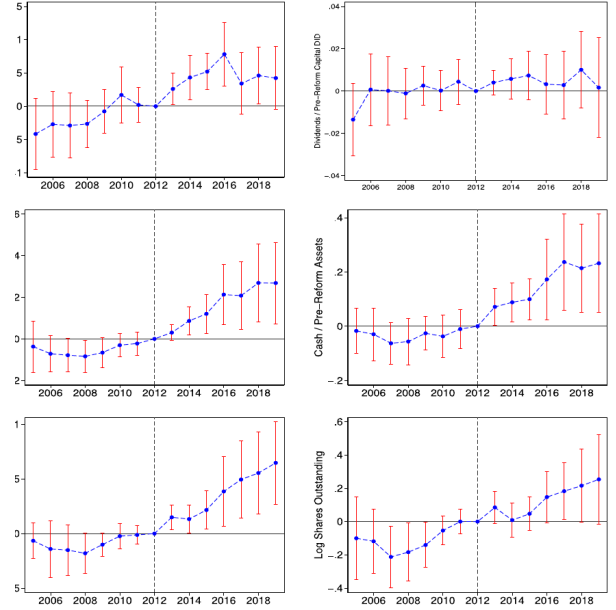
- Log shares issued increases by about 13% with controls.
- Debt relative to pre-reform assets increases by about 14 percentage points with controls.

Read: Figure 3: Dynamic plots show post-reform increases in log dividends, shares outstanding, debt, cash, and paid-in equity. Pre-trends are generally mild relative to the post-reform rise.

Economic message. The tax cut does not only change investor returns. Firms respond by distributing more cash and by increasing both equity and debt financing.

	Dependent Variable							
	Log Dividends		Dividend Dummy		Log Shares Issued		Debt	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated $\times$ Post	0.633*** (0.154)	0.403** (0.180)	0.086* (0.049)	0.056 (0.059)	0.202** (0.080)	0.133* (0.075)	0.205*** (0.053)	0.137** (0.056)
Additional Controls	No	Yes	No	Yes	No	Yes	No	Yes
Pre-Treated Mean	7.90		0.37		11.42		0.09	
Impl. Elasticity $1 - \tau_d$	0.86	0.55	0.32	0.21	0.27	0.18	3.24	2.17
Impl. Elasticity $1 - \tau_g$	1.63	1.04	0.60	0.39	0.52	0.34	6.17	4.12
Observations	1,579	1,226	2,670	1,814	2,479	1,705	2,682	1,820
Adj R-squared	0.69	0.73	0.68	0.63	0.90	0.93	0.54	0.55

(a) Table 4: dividends, share issuance, and debt



(b) Figure 3: dynamic financial-decision effects

5.4 Table 5 and Figure 4: real investment effects

Read: Table 5:

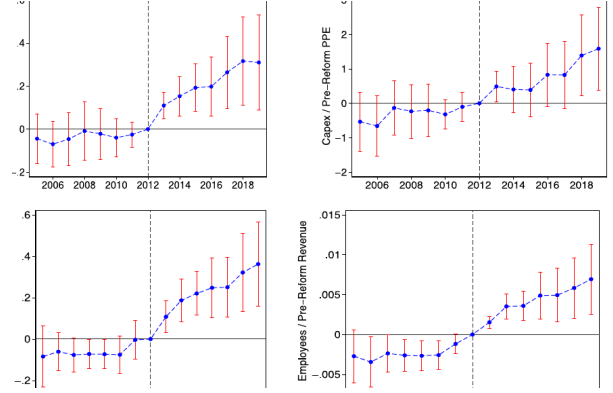
- PPE rises by about 24 percentage points relative to pre-reform assets with controls.
- Capex rises by about 43 percentage points relative to pre-reform PPE with controls.
- Total pay rises by about 8 percentage points relative to pre-reform revenue with controls.
- Employees rise by about 0.5 percentage points relative to pre-reform revenue with controls.

Read: Figure 4: The dynamic paths show clear post-reform increases in PPE, capex, pay, and employment, consistent with persistent real activity effects.

Economic message. The evidence supports the traditional view: personal tax changes can lower firms' cost of capital and stimulate real investment, not just change payout policy.

	Dependent Variable							
	PPE		Capex		Pay		Employees	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated × Post	0.251*** (0.068)	0.242*** (0.072)	1.025*** (0.361)	0.433** (0.218)	0.295*** (0.070)	0.078 (0.057)	0.007*** (0.001)	0.005*** (0.001)
Additional Controls	No	Yes	No	Yes	No	Yes	No	Yes
Pre-Treated Mean	0.18		0.52		0.21		0.01	
Impl. Elasticity $1 - \tau_d$	1.88	1.81	2.66	1.12	1.95	0.51	1.89	1.35
Impl. Elasticity $1 - \tau_g$	3.57	3.44	5.06	2.14	3.70	0.98	3.60	2.57
Observations	2,645	1,796	2,593	1,756	2,361	1,676	2,349	1,666
Adj R-squared	0.67	0.73	0.48	0.53	0.77	0.80	0.69	0.74

(a) Table 5: capital and labor investment



(b) Figure 4: dynamic real investment effects

5.5 Tables 6-7: heterogeneity and following the money

Read: Table 6:

- Cash-rich firms issue less new equity after the reform than cash-poor firms.
- Concentrated-ownership firms increase total pay more.
- Dividend-paying firms increase PPE less than non-dividend payers.

Read: Table 7:

- Money-in variables rise: paid-in equity and debt both increase.
- Money-out/held variables also rise: dividends, PPE, pay, and cash increase.
- The sums of money in and money out/held are similar, suggesting the extra funds are allocated across payouts, investment, payroll, and cash.

Economic message. The firms most likely to be financially constrained respond most strongly by raising equity and investing. The capital inflow is not simply paid out; it also finances real activity.

	Dependent Variable					
	Log Dividends	Log Shares	Debt	PPE	Pay	Employees
Panel A	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	0.478* (0.260)	0.385*** (0.117)	0.151* (0.079)	0.208* (0.117)	0.307*** (0.079)	0.006*** (0.002)
Treated × Post × Cash Rich	0.106 (0.319)	-0.361** (0.175)	0.104 (0.110)	0.052 (0.130)	-0.041 (0.139)	0.000 (0.002)
Observations	1,579	2,425	2,627	2,590	2,361	2,349
Adj R-squared	0.70	0.91	0.54	0.68	0.77	0.69
Panel B	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	0.518** (0.230)	0.174* (0.105)	0.167** (0.077)	0.238** (0.119)	0.166** (0.072)	0.004** (0.001)
Treated × Post × Concentrated Ownership	0.986* (0.511)	-0.438 (0.467)	-0.187 (0.189)	-0.389 (0.282)	0.335*** (0.095)	0.005* (0.002)
Observations	1,234	1,983	2,149	2,118	1,884	1,863
Adj R-squared	0.70	0.91	0.54	0.69	0.80	0.73
Panel C	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	-	0.356** (0.159)	0.272*** (0.081)	0.461*** (0.110)	0.401*** (0.087)	0.010*** (0.002)
Treated × Post × Dividend Paying	-	-0.225 (0.189)	-0.095 (0.096)	-0.263** (0.120)	-0.136 (0.107)	-0.004 (0.003)
Observations	-	2,479	2,682	2,645	2,361	2,349
Adj R-squared	-	0.90	0.54	0.67	0.77	0.69

(a) Table 6: heterogeneity across firms

	Dependent Variable						
	Money In		Money Out / Held				
	Paid-In Equity	Debt	Dividends	Buybacks	PPE	Pay	Cash
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated × Post	0.448*** (0.117)	0.205*** (0.053)	0.009 (0.008)	0.000 (0.001)	0.251*** (0.068)	0.135*** (0.041)	0.209*** (0.052)
Pre-Treated Mean	0.36	0.09	0.01	0.00	0.18	0.09	0.11
Impl. Elasticity $1 - \tau_d$	1.69	3.24	1.07	0.00	1.88	2.01	2.62
Impl. Elasticity $1 - \tau_g$	3.22	6.17	2.03	0.00	3.56	3.82	4.98
Observations	2,553	2,682	2,670	2,623	2,645	2,437	2,421
Adj R-squared	0.72	0.54	0.67	0.33	0.67	0.78	0.69
Σ Treated × Post	0.653		0.604				

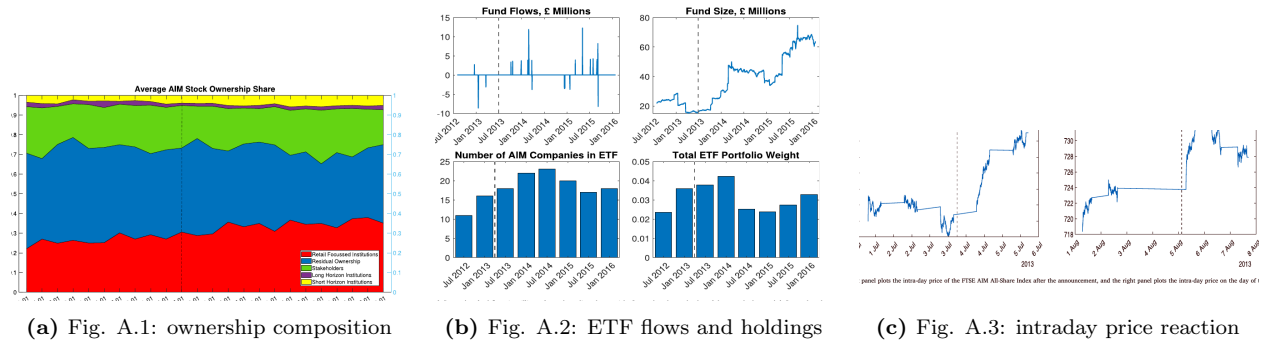
(b) Table 7: following the money

5.6 Selected appendix evidence: investor demand and robustness

Read:

- Figure A.1 shows that retail-focused institutional ownership rises after the reform, consistent with investor reallocation into AIM stocks.
- Figure A.2 shows fund-flow and ETF-level evidence that AIM-related demand increased after the legislation change.
- Figure A.3 shows intraday AIM index jumps around the announcement and implementation dates.
- Appendix tables show that the return and real-outcome results are robust to alternative factor controls, pricing outcomes, and logged outcome specifications.

Economic message. The appendix supports the core mechanism: investor demand moves toward AIM stocks, prices react immediately, and the main estimates survive alternative specifications.



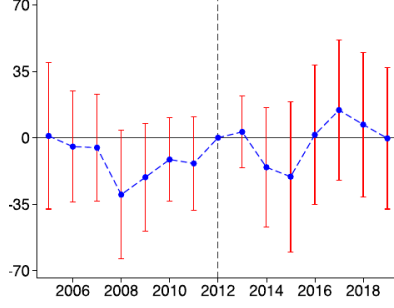
the market factor is included. Standard errors are in parentheses below coefficient estimates, and are double-clustered at the firm and the year level. The null hypothesis value for β assuming tax capitalization is stated above the correct FTSE sample as a control group, and column (2) uses the unmatched FTSE control group. Compared to column (2), column (3) uses the rate quartile from the pre-legislation period and column (4) uses the lowest quartile. Stars denote whether coefficient estimates are significant at the 1% level: ***p < 0.01, **p < 0.05, *p < 0.1.

Dependent Variable: Monthly Return				
	$H_0: \beta = -0.9\%$	$H_0: \beta = -1.9\%$	$H_0: \beta = -0.03\%$	
Panel A: All Controls	(1)	(2)	(3)	(4)
Treated × Post	-0.011 (0.004)	-0.009 (0.004)	-0.021 (0.008)	-0.002 (0.007)
Observations	23,364	85,975	77,872	77,120
Adj R-squared	0.19	0.28	0.30	0.30
Panel B: All ex. Momentum	(1)	(2)	(3)	(4)
Treated × Post	-0.014 (0.004)	-0.010 (0.004)	-0.020 (0.008)	-0.005 (0.008)
Observations	23,364	85,975	77,872	77,120
Adj R-squared	0.19	0.28	0.30	0.30
Panel C: Market Factor Only	(1)	(2)	(3)	(4)
Treated × Post	-0.015* (0.003)	-0.009 (0.004)	-0.018 (0.006)	-0.001 (0.007)
Observations	32,649	131,309	119,988	118,919
Adj R-squared	0.18	0.28	0.30	0.31

(a) Table A.1: return robustness

Dependent Variable						
	Market to Book Price	Log Turnover	Log Volume	Log Analysts		
	(1)	(2)	(3)	(4)	(5)	(6)
Treated × Post	2.405*** (0.715)	4.065*** (1.399)	-0.275 (0.563)	0.637*** (0.230)	0.594*** (0.138)	0.433*** (0.098)
Observations	2,774	1,803	1,732	29,328	32,746	28,821
Adj R-squared	0.65	0.64	0.59	0.52	0.67	0.61

(b) Table A.2: valuation and trading effects



4. This figure plots the coefficient estimates of the debt relative to contemporaneous equity dynamic difference-in-differences regression. The baseline year is 2012, t before the announcement. Standard errors are clustered at the firm level. Firm

(a) Fig. A.4: debt-equity dynamics

If the elasticity is estimated for both taxes separately with the treated $\times$ post coefficient estimate used for γ_{it} given that the d is parentheses below coefficient estimates, and are clustered at the firm level. Stars denote whether sample averages are statistically significant: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

	Dependent Variable						
	Log Paid-In Equity	Log Debt	Log PPE	Log Capex	Log Pay	Log Employees	Log Cash
Treated $\times$ Post	0.475** (0.185)	0.688*** (0.243)	1.131*** (0.241)	1.082*** (0.221)	0.952*** (0.158)	0.799*** (0.154)	0.668*** (0.207)
Additional Controls	No	No	No	No	No	No	No
Pre-Treated Mean	10.25	9.31	8.96	7.66	8.89	5.23	9.12
Impl. Elasticity $1 - \gamma_d$	0.64	0.93	1.53	1.46	1.29	1.08	0.90
Impl. Elasticity $1 - \gamma_p$	1.22	1.77	2.91	2.78	2.45	2.05	1.72
Observations	2,553	2,128	2,618	2,588	2,437	2,394	2,407
Adj R-squared	0.78	0.68	0.79	0.70	0.87	0.89	0.61

(b) Table A.5: logged outcomes, no con-

	Log Paid-In Equity	Log Debt	Log PPE	Log Capex	Log Pay	Log Employees	Log Cash
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treated $\times$ Post	0.291*** (0.147)	0.475 (0.306)	0.988*** (0.241)	0.918*** (0.249)	0.652*** (0.136)	0.733*** (0.162)	0.537*** (0.192)
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pre-Treated Mean	10.25	9.31	8.96	7.66	8.89	5.23	9.12
Impl. Elasticity $1 - \gamma_d$	0.39	0.64	1.34	1.24	0.88	0.99	0.75
Impl. Elasticity $1 - \gamma_p$	0.75	1.22	2.54	2.36	1.68	1.88	1.43
Observations	1,763	1,473	1,779	1,761	1,690	1,664	1,692
Adj R-squared	0.78	0.66	0.83	0.76	0.89	0.89	0.63

(c) Table A.6: logged outcomes, controls

6 Short summary

- The paper studies the 2013 UK reform that allowed AIM stocks to be held in tax-exempt ISAs.
- AIM stocks are treated; main-market FTSE stocks, especially matched FTSE firms, are controls.
- The pre-reform effective tax rate for AIM stocks is about 10.7% per year or 0.9% per month.
- Stock-return tests show that AIM excess returns fall by roughly this amount after the reform.
- The return effect supports tax capitalization: investors no longer require compensation for personal taxes once the asset becomes tax-exempt.
- Firm-decision tests show that AIM firms increase dividends, equity issuance, debt, PPE, capex, pay, employment, and cash.
- Dynamic plots support persistent post-reform effects and generally mild pre-trends.
- Heterogeneity shows stronger financing and investment responses among firms more likely to be constrained or non-dividend-paying.
- Investor-ownership and ETF-flow evidence support the demand mechanism.
- The broad conclusion is that personal investment taxes affect both asset pricing and real corporate decisions.

7 Conclusion

7.1 Core conclusion

Kontoghiorghe shows that eliminating personal capital gains and dividend taxes for AIM stocks inside ISAs reduced the required pretax return on AIM stocks and changed firm behavior. The estimated return decline is close to the pre-reform effective tax rate, and the corporate response includes both payout increases and real investment increases.

7.2 Economic intuition

Before the reform, AIM investors required a tax premium because capital gains and dividends were taxed. After AIM stocks became ISA-eligible, that tax premium was no longer necessary for ISA investors. Greater tax-exempt demand raised prices and reduced expected returns. Lower expected returns are the firm's lower equity cost of capital, which makes external financing and investment more attractive.

7.3 Takeaway

The paper provides clean evidence against the view that personal investment taxes are only an investor-level issue. In this setting, personal taxes shape equilibrium stock returns and firm-level decisions over capital structure, payouts, capital expenditure, payroll, and employment.